What Happens When You Hit Enter

I mapped what happens behind the screen when you type a web address, e.g. google.com or bbc.co.uk. Here is each stage, what can go wrong, and how defenders stop it:

1. DNS lookup

The browser asks which IP address belongs to the name.

ATTACK DNS spoofing and cache poisoning

RISK You are sent to a fake site without knowing.

FIX DNSSEC, DoH/DoT

2. TCP connection

A connection opens with SYN -> SYN-ACK -> ACK.

ATTACK SYN flood

RISK The server runs out of connections and goes down.

FIX SYN cookies, rate limiting, DDoS protection

3. TLS handshake

The browser and server agree on encryption and the browser checks the certificate.

ATTACK Man-in-the-middle, forged certificates, downgrade attacks

RISK Someone reads or changes your traffic.

FIX Certificate validation, TLS 1.3, HSTS

4. HTTP request

The browser asks for the page, sending headers and cookies.

ATTACK Sniffing and session hijacking

RISK Stolen cookies mean stolen logins.

FIX HTTPS everywhere, Secure cookies

5. Server side

The request passes through load balancers and a WAF to the app.

ATTACK Injection, XSS, bot abuse

RISK Data theft and account takeover.

FIX WAF, input validation, rate limiting

6. HTTP response

The server sends back the page and its security headers.

ATTACK Clickjacking and script injection

RISK Weak or missing headers leave the door open.

FIX CSP, X-Frame-Options, HttpOnly cookies

7. Rendering

The browser loads scripts, images and third-party content.

ATTACK Malicious third-party scripts

RISK One compromised script can hit every visitor.

FIX Browser sandbox, same-origin policy, CSP

Key takeaway: every layer leaves evidence in the logs, and every layer can be attacked. Knowing what normal looks like is what makes the abnormal stand out.